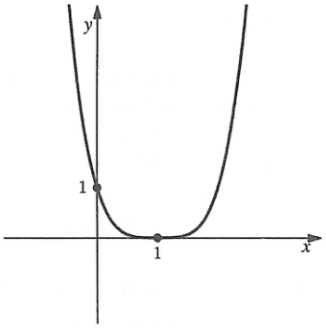
Graph Sketching: Quartic Graphs

Questions 1–15

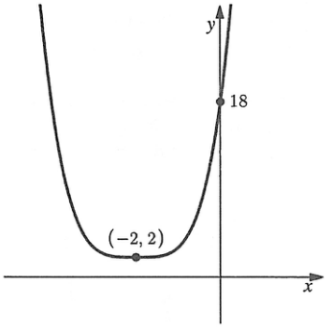
Question 1. The following graphs are all functions that can be written in the form $y = (x - p)^4 + q$. Find the equation of the curve in the form

$$y = x^4 + bx^3 + cx^2 + dx + e$$

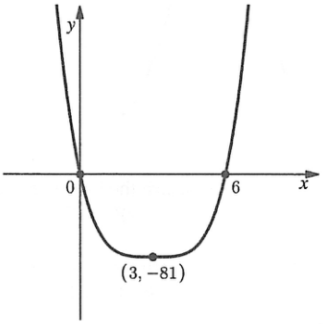
a)



b)



c)



Question 2. Sketch the following graphs:

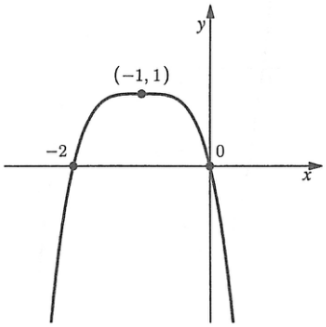
a) $y = (x - 2)(x - 1)(x + 1)(x + 2)$

b) $y = (x - 3)(x + 1)(x - 1)(x + 4)$

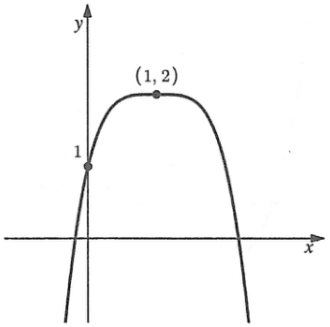
Question 3. The following graphs are all functions that can be written in the form $y = q - (x - p)^4$. Find the equation of the curve in the form

$$y = -x^4 + bx^3 + cx^2 + dx + e$$

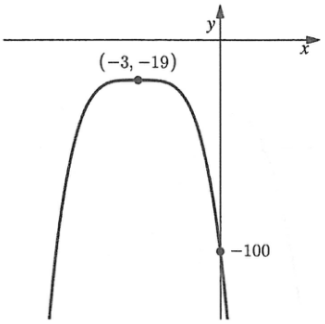
a)



b)



c)



Question 4. Sketch the following graphs:

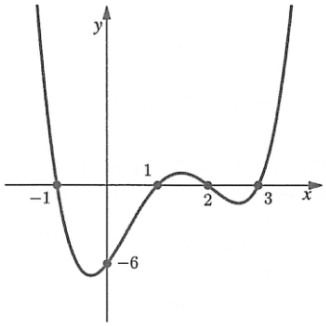
a) $y = -(x - 3)(x + 1)(x + 2)(x + 3)$

b) $y = -(x + 4)(x + 3)(x + 2)(x + 1)$

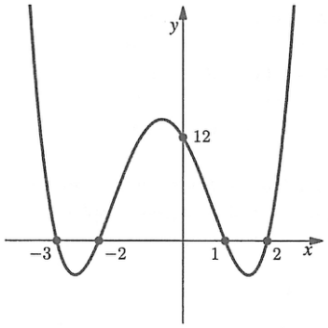
Question 5. Find the equation of each of the following graphs in the form

$$y = (x - a)(x - b)(x - c)(x - d)$$

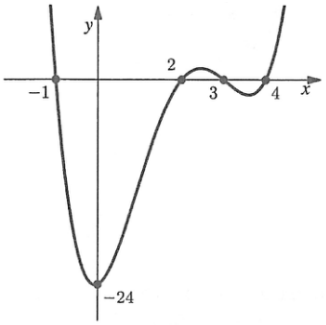
a)



b)



c)



Question 6. Sketch the following graphs:

a) $y = (x - 1)(x + 3)(x - 2)^2$

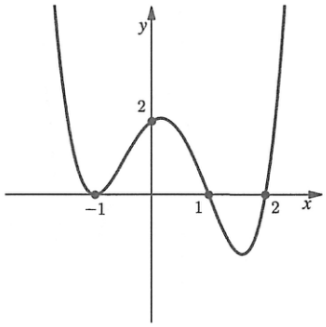
b) $y = (x + 1)^2(x + 2)(x - 3)$

c) $y = (x + 1)^2(x - 2)^2$

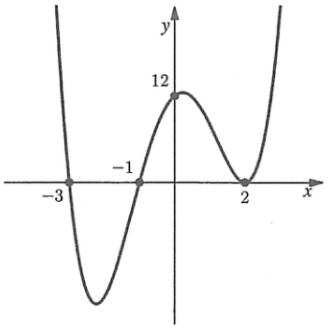
Question 7. Find the equation of each of the following graphs in the form

$$y = (x - a)(x - b)(x - c)^2$$

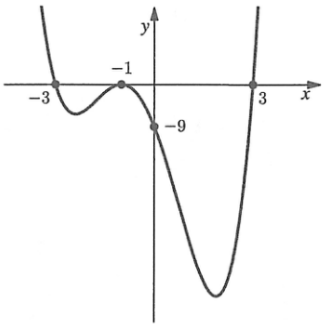
a)



b)



c)



Question 8. Sketch the following graphs:

a) $y = -(x - 4)(x + 1)(x + 2)^2$

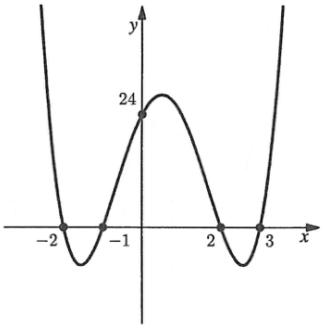
b) $y = (x - 1)(3 - x)(x + 1)^2$

c) $y = -x^2(x - 3)^2$

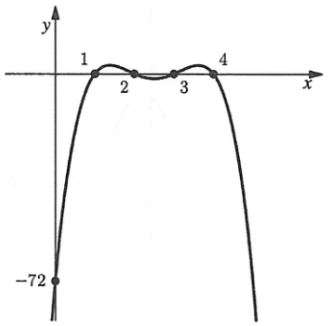
Question 9. Find the equation of each of the following graphs in the form

$$y = k(x-a)(x-b)(x-c)(x-d)$$

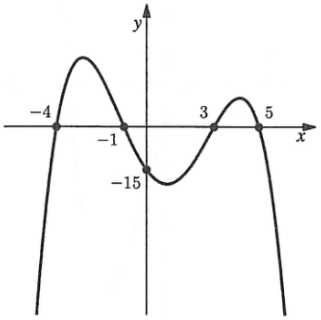
a)



b)



c)



Question 10. Sketch the following graphs:

a) $y = \frac{1}{2}(x-4)(x+2)(x-2)(x-6)$

b) $y = -\frac{2}{3}(x+3)(x-3)(x-6)^2$

Question 11. Sketch the following graphs, showing where they intersect the coordinate axes:

a) $y = (2x-1)(x+1)(2x+1)(3x-1)$
b) $y = (x+2)(2x-1)(3x-5)^2$
c) $y = x(3x+4)(5x+2)^2$
d) $y = -x^2(4x-3)(3x-4)$
e) $y = (2x-1)^2(2x+3)^2$
f) $y = -\frac{1}{6}(6x-1)^2(12x+5)^2$

Question 12. Sketch the following graphs, showing where they intersect the coordinate axes:

a) $y = (x-2)(x+3)^3$
b) $y = (2x+3)(4x-3)^3$
c) $y = (3-2x)x^3$
d) $y = -\frac{1}{4}(3x+4)(3x-1)^3$

Question 13. Given that $k > 0$, sketch:

a) $y = x(x-k)(x-2k)(x-3k)$

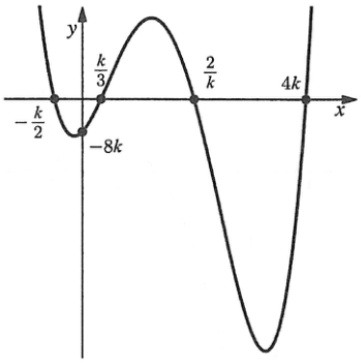
b) $y = (x+k)(x+3k)(x-2k)(x+2k)$

c) $y = (k-x)(2k+x)(3k-x)^2$

d) $y = (4k-x)(2k+x)^3$

Question 14. Given that k is a positive constant, find the equation of the following graph in the form

$$y = a(bx+c)(dx+e)(fx+g)(hx+i)$$



Question 15. Given that $k > 0$, sketch the graphs of

$$y = (x - 3k)(x + 3k)(x - k)^2$$

and

$$y = (3k - x)(3k + x)(k - x)$$

on the same axes, showing clearly where the two graphs cross the coordinate axes and where they intersect.